Chapter 5 | Written solutions

Multiple Heads, Position, and the Complete Architecture

All fifteen exercises are solved here in the order of the chapter. We follow one prediction from its permitted inputs to its vocabulary scores, work through small numerical examples, and test the complete model's information paths. Make a prediction before running each experiment. The written PDF contains every explanation; the notebook PDF also includes the code and its recorded results.

1. Follow one prediction

To predict *coche*, the decoder receives BOS and *el* at positions 0 and 1. The correct answer *coche* is the label at position 1 and the input at the later position 2. The causal mask blocks that later input. The entire source *the red car* is available, along with its end marker when present.

Operation	Information used	Result at this target position
Target self-attention	Available target prefix	A mixture of target value vectors
Cross-attention	Query from the updated target; keys and values from final source memory	A mixture of source value vectors
FFN	Features of this one updated row	A transformed feature vector
Vocabulary projection	Features of the final decoder row	One logit per vocabulary entry

The two attention operations communicate between positions. The FFN and the vocabulary projection act on each row independently, using shared parameters across rows. Residual additions and LayerNorm also act within a row. Cross-attention's softmax distributes weight over source keys. The final softmax distributes probability over vocabulary entries. A source word is a place to read information; a vocabulary entry is a possible prediction.

The masks permit reading *car* while predicting *coche*, even though these words have different positions in their sentences. Permission does not mean an untrained head has learned that alignment.

2. Follow the head axes

Start with one example and two token rows. The first token carries $(1, 2, 3, 4)$ and the second carries $(5, 6, 7, 8)$. We divide model width 4 among two heads, so the width per head is $4/2 = 2$. Within each token, head 1 gets its first two coordinates and head 2 gets its last two.

Head, counted from 1	Token 1	Token 2
1	$(1, 2)$	$(5, 6)$
2	$(3, 4)$	$(7, 8)$

The input axes are `[batch, token, model feature]`. Reshape the last axis of length 4 into two axes of length 2. The result has axes `[batch, token, head, head feature]`. At this stage, indexing the second axis still selects a token and gives both of that token's heads. Transpose the token and head axes to obtain `[batch, head, token, head feature]`. Now indexing the second axis gives the two tokens for one head, as shown in the table.

Using zero-based indices b for example, h for head, t for token, and j for head feature, the element at $[b, h, t, j]$ in the split tensor is the input at $[b, t, 2h + j]$. In particular, head 1's second token takes the first two coordinates of $(5, 6, 7, 8)$.

To merge, transpose the token and head axes back. Each token now holds its two head pieces consecutively. Reshape the head and head-feature axes into one width-four feature axis. This should recover the input entries in their original order.

The two layouts have the same numerical shape because the token and head axes both have length two. To see what a shape check would miss, pass the reshape-only tensor to `merge_heads` as though its second axis were the head axis. Its first two rows under that interpretation would be `[1, 2]` and `[3, 4]`, two pieces of the first token. That incorrect grouping sends different entries into the merge.

3. Project before splitting

The added entry in row 4, column 1 of the row-convention matrix makes the first projected feature depend on both the first and fourth input features:

$$(x_1, x_2, x_3, x_4)W^Q = (x_1 + x_4, x_2, x_3, x_4).$$

For $(1, 2, 3, 4)$ the projected row is $(5, 2, 3, 4)$, so the two heads receive $(5, 2)$ and $(3, 4)$. Replacing the last input feature by 9 gives $(10, 2, 3, 9)$; head 1 now receives $(10, 2)$ and head 2 receives $(3, 9)$. Head 1 changes even though the original first two input features stay fixed.

A head owns a subset of the *projected* coordinates. Each of those coordinates can depend on all input features through its learned projection weights. The transpose used below is essential: `nn.Linear` stores output features as rows, while the book writes input features as rows of the weight matrix.

4. Compare three attention shapes

An attention map has axes `[batch, head, query, key]`. First imagine a single example and a single head. Each query needs one score for every stored key position, so it contributes one row. Stacking heads and then examples adds the first two axes.

Operation	Query rows	Key columns	Full shape
Encoder self-attention	5 source positions	5 source positions	$[2, 4, 5, 5]$
Decoder self-attention	3 target positions	3 target positions	$[2, 4, 3, 3]$
Cross-attention	3 target positions	5 source positions	$[2, 4, 3, 5]$

Softmax normalizes over the final axis for each example, head, and query separately. A mask changes which entries may receive weight without changing the stored map shape. It does not normalize over heads or query positions.

A triangle is meaningful when both axes follow the target timeline: a query may read its own and earlier inputs. Applying the same condition $j \leq i$ to cross-attention would let the first target query read only the first source position. In this example it would prevent every target query from reading source positions 4 and 5. Translation does not require word-by-word alignment in the same order. The full source is already available.

Instead, a source-validity mask of shape $[2, 1, 1, 5]$ excludes source padding and broadcasts over heads and queries. It works for encoder self-attention as well as rectangular cross-attention. The mask's size-one query axis means every query sees the same set of valid source keys.

5. Move a token

The position index starts at zero, and trigonometric arguments are in radians. At width four there are two sine/cosine pairs. The first uses divisor $10000^0 = 1$; the second uses $10000^{1/2} = 100$. At position 2,

$$\mathrm{PE}_2 = (\sin 2, \cos 2, \sin 0.02, \cos 0.02) \approx (0.909297, -0.416147, 0.019999, 0.999800).$$

The same token keeps its embedding $\mathbf{e} = (0, 1, 0, 2)$. Multiply it by $\sqrt{4} = 2$ before adding the position vector:

$$\mathbf{u}_2 = (0, 2, 0, 4) + \mathrm{PE}_2 \approx (0.909297, 1.583853, 0.019999, 4.999800).$$

At position 1 the angles are 1 and 0.01. The resulting input is $\mathbf{u}_1 \approx (0.841471, 2.540302, 0.010000, 4.999950)$. The second coordinate changes substantially because $\cos 1$ and $\cos 2$ differ substantially. The last coordinate changes much less because its angle changes by only 0.01.

The embedding, its scale factor, the frequency divisors, and the final width stay fixed. Only the position index changes. Subtraction cancels the common embedding term: $\mathbf{u}_2 - \mathbf{u}_1 = \mathrm{PE}_2 - \mathrm{PE}_1$. These are the inputs before dropout. The division by $\sqrt{d_k}$ belongs to a later operation on attention scores.

6. Inspect a post-norm wrapper

The wrapper applies dropout to the sublayer's update, adds that result to the incoming vector, and normalizes the sum. Here the update is zero, so the dropout output is zero. For incoming vector $\mathbf{x} = (1, 2, 3, 4)$, the residual sum is $\mathbf{r} = \mathbf{x} + (0, 0, 0, 0) = (1, 2, 3, 4)$.

LayerNorm works across the four features of this one vector. Its mean is $\mu = (1 + 2 + 3 + 4)/4 = 2.5$. The centered features are $(-1.5, -0.5, 0.5, 1.5)$. Their squared values are $(2.25, 0.25, 0.25, 2.25)$, so the population variance is $\sigma^2 = (2.25 + 0.25 + 0.25 + 2.25)/4 = 1.25$. The divisor counts all four features.

With scale one and offset zero at each feature, the answer is

$$\frac{(-1.5, -0.5, 0.5, 1.5)}{\sqrt{1.25 + 10^{-5}}} \approx (-1.341635, -0.447212, 0.447212, 1.341635).$$

The residual addition carried the input into the sum. The normalization that follows centers and rescales that sum, so the whole wrapper changes the vector even with zero update.

For incoming vector $(3, 3, 3, 3)$, the residual sum is $(3, 3, 3, 3)$, its mean is 3, and every centered feature is zero. The variance is zero. The denominator is $\sqrt{10^{-5}} > 0$, so division produces a zero vector and the affine transformation with scale one and offset zero leaves it at $(0, 0, 0, 0)$. The added constant keeps the computation defined in this case.

7. Track dropout values

During training with drop probability $p = 1/2$, a coordinate x_j is dropped with probability $1/2$. If it is retained, the layer divides it by the keep probability $1 - p = 1/2$. Write $M_j = 0$ or 1 for the random choice. The output is $M_j x_j/(1 - p) = 2M_j x_j$.

Input coordinate	Dropped, probability $1/2$	Retained, probability $1/2$	Expected output
1	0	2	$\frac{1}{2}(0)+\frac{1}{2}(2)=1$
2	0	4	$\frac{1}{2}(0)+\frac{1}{2}(4)=2$
3	0	6	$\frac{1}{2}(0)+\frac{1}{2}(6)=3$
4	0	8	$\frac{1}{2}(0)+\frac{1}{2}(8)=4$

The expected vector is $(1, 2, 3, 4)$. That average describes many possible masks. One training call yields one sample, with each coordinate either zero or twice its input. In evaluation mode, dropout returns $(1, 2, 3, 4)$ on every call.

The following check keeps the module in training mode for two calls. It resets the CPU random seed to produce the same mask in both. The first call records gradients for the floating-point input, and the second runs inside `torch.no_grad()`. The input requests gradients so the output flags distinguish those two contexts. The final call switches the module to evaluation mode and runs with gradient recording enabled.

Each saved training coordinate is either zero or twice its input. Training mode controls the dropout rule. Gradient recording controls whether PyTorch tracks differentiable operations. Both training calls sample and rescale values, even though only one records gradients. Evaluation switches dropout to its identity rule, and the evaluated result retains gradient tracking through the input. These two controls answer different questions.

8. Follow both heads.

Start with the second input row, $(0, 1, 0, 2)$. The query and key projections are identities. Head 1 receives the first two projected coordinates and head 2 receives the last two. The value projection swaps those two coordinate pairs, so head 2 reads the first pair from each input row as its value.

For head 2 the second query is therefore $(0, 2)$. Write its keys and values in the same token order as the input. The key and value on a row refer to the same position.

Position	Head 2 key	Head 2 value	Dot product with $(0, 2)$
1	$(1, 0)$	$(1, 0)$	$0(1)+2(0)=0$
2	$(0, 2)$	$(0, 1)$	$0(0)+2(2)=4$
3	$(1, 1)$	$(1, 1)$	$0(1)+2(1)=2$

The key width of one head is two, so the scaled scores are $(0, 2\sqrt{2}, \sqrt{2})$. Define $r = \exp(\sqrt{2})$ and $D_2 = 1+r^2+r$. The second row of head 2's attention map is

$$\left(\frac{1}{D_2}, \frac{r^2}{D_2}, \frac{r}{D_2}\right) \approx (0.045388,\ 0.767918,\ 0.186694).$$

Each weight scales the value at its key position. The three scaled values are $(1/D_2, 0)$, $(0, r^2/D_2)$, and $(r/D_2, r/D_2)$. Summing each feature separately gives

$$\mathbf{o}_2^{(2)} = \left(\frac{1+r}{D_2}, \frac{r^2+r}{D_2}\right) \approx (0.232082,\ 0.954612).$$

The most heavily weighted value is $(0, 1)$ at position 2. This contributes much of the second coordinate and contributes zero to the first.

Head 1's second query is $(0, 1)$. Against its keys $(1, 0), (0, 1), (1, 1)$ the raw score row is $(0, 1, 1)$. Let $u = \exp(1/\sqrt{2})$ and $D_1 = 1 + 2u$. The weight row is $(1, u, u)/D_1$. Its values are $(1, 0), (0, 2), (1, 1)$, so the read is $((1 + u)/D_1, 3u/D_1) \approx (0.598888, 1.203336)$.

Both reads belong to query position 2. Concatenate their feature coordinates in head order to obtain $(a, b, c, d) \approx (0.598888, 1.203336, 0.232082, 0.954612)$. The output projection maps the row to $(a + c/2,\ b + d/2,\ a/2 + c,\ b/2 + d)$. Using full-precision values gives

$$(0.714929,\ 1.680642,\ 0.531526,\ 1.556280) \quad \text{to six decimal places.}$$

The output projection combines features at this one position. The weighted sums inside the heads combined values across positions.

The returned attention tensor has shape `[1, 2, 3, 3]`. Its axes are batch, head, query position, and key position. In code, `weights[0, 1, 1]` therefore selects all key weights for head 2 at query position 2. The returned concatenated and projected outputs both have shape `[1, 3, 4]`, with axes batch, query position, and feature.

The first raw score row happens to be $(1, 0, 1)$ in both heads. In head 1 and head 2 the first query is $(1, 0)$ and the first key coordinates across positions are $1, 0, 1$. This makes the first weight rows identical. The second query gives $(0, 1, 1)$ in head 1 and $(0, 4, 2)$ in head 2. Normalizing those rows gives different distributions. Each row describes one query's comparisons, so agreement at the first position says nothing about the later queries.

9. Separate concatenation from mixing.

Call `toy_multihead(output_projection=torch.eye(4))` to use the identity for W^O. The queries, keys, and values still come from the same input and the same three projection matrices. This gives the original attention weights and the original head reads. Concatenation places the head reads side by side at each position, so that tensor is also unchanged. Multiplying the concatenation by the identity leaves it as is. Of the three values returned by the helper, only the projected output changes.

In PyTorch, a bias-free `nn.Linear` stores its weight with axes output feature, input feature. The matrix used for row-vector multiplication in the book has axes input feature, output feature. Copy the transpose of the book's value matrix into `v_proj.weight`. The two identity matrices for queries and keys and the new identity for the readout all equal their own transposes.

Fix the model width at eight. With two heads, each head gets four output features from each input projection. The query projection can be viewed as two 8×4 matrices side by side, giving $2(8)(4) = 64$ parameters. Keys and values each add 64. With four heads, the query projection is four 8×2 matrices side by side and still has $4(8)(2) = 64$ entries. The same holds for keys and values.

Concatenation yields eight features in both cases. The output projection mixes them into an eight-feature output and has $8(8) = 64$ entries. All four projections are bias-free.

Heads	Width per head	Query	Key	Value	Output	Total
2	4	64	64	64	64	256
4	2	64	64	64	64	256

Changing the number of heads changes how the projected features are partitioned and how many attention maps are computed. It leaves the full projection sizes fixed when the total model width is fixed.

10. Test the positionwise FFN.

In the toy feed-forward network, a row first expands from width four to eight. With zero bias, its first four entries are the original row and its last four entries are their negatives. For $x = (1, -2, 0, 3)$ this gives $(1, -2, 0, 3, -1, 2, 0, -3)$. ReLU replaces negative entries with zero, giving $(1, 0, 0, 3, 0, 2, 0, 0)$.

The second linear map adds the first group of four to the second group coordinate by coordinate. It gives $(1, 2, 0, 3)$. For any coordinate x_i of an input row, the resulting function is $\max(0, x_i) + \max(0, -x_i) = |x_i|$ for these particular weights.

Construct a fresh module. The expansion stores $[I_4; -I_4]$ as an 8×4 weight array and the contraction stores $[I_4\ I_4]$ as a 4×8 weight array, since the output feature axis comes first in the stored weights. The canonical helper returns the numbers for a single input and can check our work.

Put three different vectors in a tensor of shape `[1, 3, 4]`. The axes index the batch item, token position, and features. Both linear maps operate on the final feature axis, and ReLU operates on each entry. There is no reduction over the token axis. Changing the token order therefore changes the output order in the same way. Replacing just token 2 leaves the outputs for tokens 1 and 3 as they were.

11. Read a vocabulary distribution.

Use the separate toy decoder row $y = (1, -1, 0, 2)$ and zero bias. Each logit is the dot product of y with one column of the vocabulary weight matrix. The vocabulary order is *el*, *coche*, EOS.

Token	Column	Dot product with y	Logit
el	$(1, 0, 0, 1/2)$	$1(1) + (-1)(0) + 0(0) + 2(1/2)$	2
coche	$(0, -1, 0, 0)$	$1(0) + (-1)(-1) + 0(0) + 2(0)$	1
EOS	$(-1, 0, 1, 1/2)$	$1(-1) + (-1)(0) + 0(1) + 2(1/2)$	0

Exponentiating produces $(e^2, e, 1)$. Their sum $D = e^2 + e + 1$ normalizes the three vocabulary entries. The probability row is $(e^2/D, e/D, 1/D)$, approximately $(0.665241, 0.244728, 0.090031)$.

The loss for the reference word *coche* uses its probability e/D, before rounding. The negative natural logarithm is

$$-\log(e/D) = \log D - 1 = \log(e + 1 + e^{-1}) \approx 1.407605964444380.$$

The logits for this single prediction have shape `[1, 3]`. The first axis counts the one prediction row and the final axis indexes the vocabulary. The target is a length-one tensor containing index 1, the zero-based index of *coche* among these three classes. There is no source-position or head axis in this tensor. `F.cross_entropy` accepts logits because it performs the log normalization internally.

12. Count the predictions that matter

The first example has four labels and the second has three real labels plus one PAD label. Thus seven of the eight stored rows contribute to the loss. With all twelve logits equal to zero, each vocabulary entry receives $1/12$. Each valid label therefore contributes $-\log(1/12) = \log 12$. The ordinary, unweighted mean cross-entropy is

$$\mathcal{L} = \frac{7 \log 12}{7} = \log 12 \approx 2.484907.$$

With `reduction="none"`, PyTorch returns zero at the ignored row. Taking the mean of all eight returned values divides by eight and gives $(7/8)\log 12 \approx 2.174293$. Zeroing an excluded term and choosing the correct denominator are separate operations.

The loss is independent of all logits at the ignored row, so every derivative there is zero. At a valid row its derivative with respect to logit z_j is $(p_j - \mathbf{1}[j = y])/7$. The factor $1/7$ comes from averaging the seven scored predictions. PAD remains one of the twelve competing vocabulary entries at these valid rows; ignoring a label does not delete a vocabulary class. Its derivative there is $(1/12)/7 = 1/84$.

13. Keep query validity separate from key validity

In the shorter source sentence, the last two entries are PAD. Those entries occupy space in the batch so that both sentences have width five. Change the IDs in the padding slots and keep their validity flags fixed. If a real target position is insulated from those slots, all twelve of its vocabulary logits should agree before and after the change.

The code recreates the one-layer demonstration with seed 7, model width four, two heads, and FFN width eight. It uses the same twelve-entry source and target vocabularies. Evaluation mode disables dropout. The two forward passes in this exercise only inspect values, so they also run with gradient recording off.

The saved Boolean source mask marks **key slots**. The encoder uses it at every source query, and the decoder uses it for every cross-attention query. In either place the mask broadcasts from shape $[B, 1, 1, S]$ over heads and queries. Disallowed key scores are set to $-\infty$ before softmax, so their weights are zero.

A padded source query can still produce a hidden vector. In an encoder layer, its vector remains in a masked key slot in the next layer. The decoder also masks that slot when it reads the final encoder memory. Residual connections, LayerNorm, and FFNs operate within each token row, so these sublayers cannot move the padded vector into a valid source row. Recomputing validity from the edited IDs would mark ID 11 as real and change the experiment. Pass the original flags to both forward passes.

The batch has seven valid target queries. The last query of the second example occupies a padded slot. Its decoder self-attention can still read the three valid target keys that precede it. Its cross-attention can also read the three real source keys. The final linear projection produces a logit for each vocabulary entry at this slot as it does at a real query.

The label at that padded query is PAD, ID 0. The training loss uses `ignore_index=0`, so the query contributes neither a loss term nor one of the positions in the mean. For the seven remaining labels, cross-entropy still normalizes over the whole target vocabulary. A key mask controls what a query can read. The label determines whether the loss scores its output. In this particular batch the valid input slots and the scored label slots coincide.

Verify that filtering to seven rows before computing the loss gives the same result as flattening all eight rows and using `ignore_index`. Then assign a different vector of logits to the ignored row and compare the loss again.

For the checked seed-7 model, changing only the source padding leaves all 84 scored logits (seven rows of twelve) unchanged. The padded target row is nonzero: for example, its first logit is approximately 0.3070. The loss is 2.973556394290 whether we use `ignore_index`, filter to the seven real labels, or replace every logit at the ignored target row. Its values are irrelevant to this objective, even though the model computes them.

14. Break causality deliberately (optional challenge)

The first decoder input is `[1,8,9,10]`. Change only its third entry, from 9 to 6. With a causal mask, the first two queries cannot read this entry. Their logits must remain unchanged in evaluation mode. Removing the triangle opens that information path, while keeping the key-validity mask still excludes padding.

We temporarily replace the module's `build_masks` function, using a context manager that restores it automatically. The replacement calls the original function to obtain the source mask, then returns target key validity without the triangle. The same model and weights are used for every comparison; only the target mask changes. The canonical source file is never edited.

First measure the safe model. Then temporarily substitute the keys-only mask and measure the same change again. Summarize each of the first two logit vectors by its maximum absolute change. The assertions check the complete vectors for the causal model and require a measurable change in both positions for the chosen noncausal example.

Changing the third input changes its projected key and value. Without the triangle, an earlier query can change its weight distribution and read a different value mixture. That change propagates into vocabulary logits. Restoring the triangle closes the future-to-past route.

The companion helper also compares a prefix by itself with that prefix inside the complete target. The noncausal version fails because the longer input supplies additional readable keys. After the context ends, the same model passes again. A missing causal mask creates a possible leak; the size of an observed change depends on the weights and inputs.

In the checked run, changing the future input changes the first two noncausal logit vectors by approximately 0.2736 and 0.4676 in maximum absolute value. Both changes are zero with the causal mask. The prefix helper rejects the keys-only mask and passes after restoration. These numbers describe this initialization; the invariant to preserve is the absence of future influence.

15. Build a deeper model (optional challenge)

A stack of two layers applies two transformations in sequence. Each layer has its own learned parameters. When a parameter has the same name in both layers, compare their Python identities and their first-element addresses using `data_ptr()`. In this model the corresponding arrays are separate tensors in separate storage allocations.

A LayerNorm scale starts at one in every coordinate and its offset starts at zero. Equal initial values describe the entries. The identity and storage comparisons establish that changing one layer's parameter need not change the other's.

The encoder first replaces its hidden array once per layer. The second encoder layer's output becomes the final memory, with shape $[2, 5, 4]$. The decoder loop updates the target array at every layer and passes that same memory tensor to every layer. Each cross-attention module uses its own learned projections to turn the shared memory into keys and values. The queries come from the decoder's current target states.

Temporary hooks observe the tensors that enter and leave these modules during the forward pass. Capture the final encoder output, then check that both decoder layers' cross-attention modules receive that very object as their key and value inputs. The hooks observe the tensors without modifying the computation.

The forward pass was in evaluation mode and outside `torch.no_grad()`. This leaves autograd active and keeps dropout disabled. Flatten the batch and target axes, compute mean cross-entropy over the seven non-PAD labels, and call `backward()`. Check that each parameter receives a finite gradient. A nonzero gradient in the source embeddings gives a direct check that this target loss can flow back through cross-attention and the encoder.

With seed 7, the two-layer model passes prefix and padding checks and has masked loss approximately 2.474837. All 64 parameter tensors receive finite gradients: 24 in the encoder stack (12 per layer), 36 in the decoder stack (18 per layer), two embedding tables, and two vocabulary-projection tensors. The sum of absolute source-embedding gradient entries is approximately 0.065284. These results confirm a connected backward path, not successful learning.